

Auditable Editing for Structured Research Papers

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Abstract

Research editors must balance structured interaction with exact source preservation. We describe a synthetic benchmark for small, revision-checked LaTeX edits and report illustrative results on a deliberately modest corpus. The example demonstrates Setwright’s supported paper structures; it does not make an empirical product claim.

1 Introduction

LaTeX gives researchers durable, inspectable source, while structured editors can reduce the cost of common operations. Existing scientific workflows also depend on citations, stable cross-references, and mathematical notation [1, 2]. A dependable bridge between those interfaces cannot silently rewrite syntax it does not understand.

We study a narrow question: can a structured edit be represented as a minimal byte patch and rejected when its source precondition is stale? This sample uses a static include graph so every visual operation remains inside one file boundary. A source span is measured in UTF-8 bytes, not displayed characters.¹

Our illustrative pipeline is shown in figure 1; the validation rule is formalized in section 2.

2 Method

Definition 1 (Validated edit). *A validated edit is a tuple $e = (f, b_0, b_1, r, h)$ containing a file identifier, half-open byte interval, replacement text, and the expected hash of the replaced bytes.*

Let S_f be the authoritative bytes for file f . An edit is eligible exactly when its interval is valid and its precondition matches:

$$0 \leq b_0 \leq b_1 \leq |S_f| \quad \wedge \quad H(S_f[b_0 : b_1]) = h. \quad (1)$$

Theorem 1 (Stale-edit exclusion). *If the bytes inside an edit’s interval change after the edit is proposed, a collision-resistant expected-slice hash causes validation to reject the edit, except with negligible collision probability.*

Proof. Changing the interval changes the hash input. Acceptance would therefore require a collision or second preimage for the selected digest. $\square$

The benchmark applies an edit, reparses the candidate, checks complete source coverage, and then applies the inverse edit. A compact representation is:

¹This distinction matters for non-ASCII source text.

Intent $\rightarrow$ minimal patch $\rightarrow$ candidate parse $\rightarrow$ atomic commit

Figure 1: A source-authoritative editing pipeline. The diagram is rendered with ordinary LaTeX so the sample has no binary asset dependency.

Table 1: Synthetic edit-validation outcomes.

Fixture class	Accepted	Correctly rejected
Plain paragraphs	120	0
Equations	84	6
Raw environments	0	32
Stale source spans	0	40

Listing 1: Illustrative edit loop.

```
for edit in edits:
    candidate = validate_and_apply(source, edit)
    assert complete_coverage(candidate)
    source = apply_inverse(candidate, edit)
```

3 Illustrative results

Table 1 contains synthetic values chosen only to exercise table editing and references. They are not measurements of the Setwright scaffold.

The intended interpretation follows equation (1): ordinary edits with matching preconditions proceed, while stale or intentionally unsupported edits remain explicit. A real evaluation would report corpus licenses, hardware, software versions, confidence intervals, and failure cases.

4 Limitations and conclusion

The benchmark omits arbitrary macro expansion and stateful TeX. Those regions must remain exact raw source rather than being normalized into a visual model. Future work should evaluate larger, permissively licensed paper corpora and all supported operating-system sandboxes.

References

- [1] Leslie Lamport. *LaTeX: A Document Preparation System*. Addison-Wesley, 2 edition, 1994.
- [2] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.